



CDA 3103

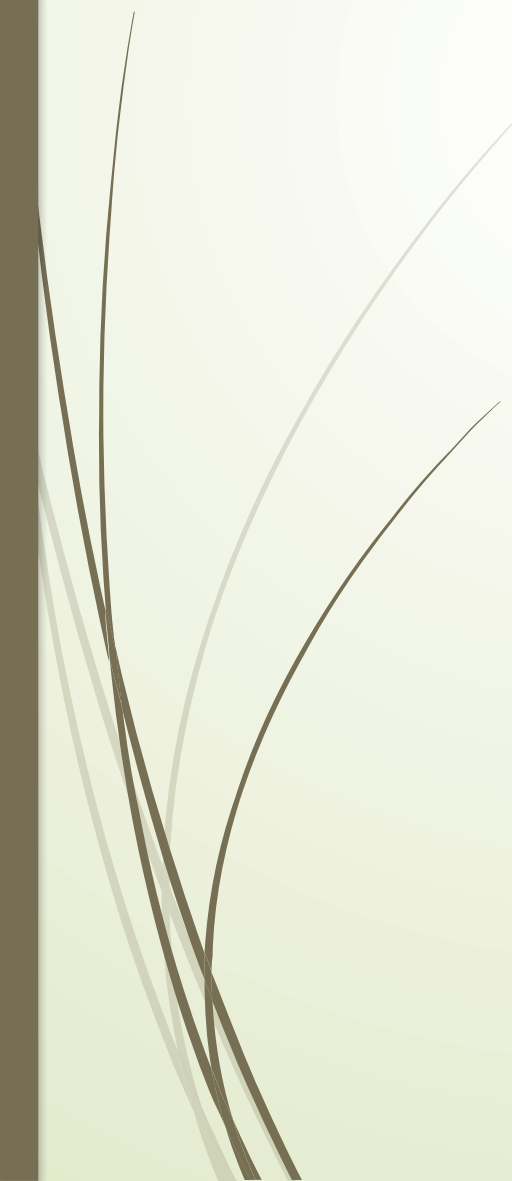
Recitation 1 – Number Conversions

Converting: Binary to Decimal

- Express the binary number, n , in the form: $a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$
- Use 2 for b (the base is 2)
- Fill in k
 - Start with 0 for the least significant bit and count upward
- Fill in a_k
 - These will be 0s or 1s
- Solve the equation



Converting: Decimal to Binary

- Continually divide the decimal number by 2, recording the remainder each time, until 0 is reached. The remainders (read upwards from bottom to top) are the binary result.
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Question 1

- Convert $(11101011)_2$ to decimal.
- $a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$
- $1*2^7 + 1*2^6 + 1*2^5 + 0*2^4 + 1*2^3 + 0*2^2 + 1*2^1 + 1*2^0$
- $128 + 64 + 32 + 0 + 8 + 0 + 2 + 1$
- 235_{10}

Question 2

Convert $(937)_{10}$ to binary.

$937 / 2 = 468$ R 1

$468 / 2 = 234$ R 0

$234 / 2 = 117$ R 0

$117 / 2 = 58$ R 1

$58 / 2 = 29$ R 0

$29 / 2 = 14$ R 1

$14 / 2 = 7$ R 0

$7 / 2 = 3$ R 1

$3 / 2 = 1$ R 1

$1 / 2 = 0$ R 1

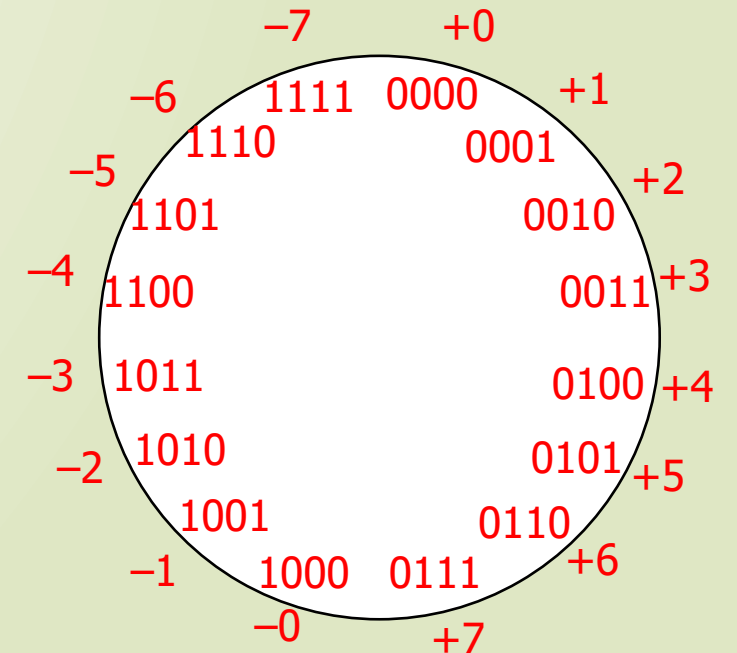
1110101001_2

Signed Numbers: Sign and Magnitude Method

- The most significant bit is dedicated to sign
 - sign: 0 = positive (or zero), 1 = negative
- Other bits represent the absolute value or magnitude
 - three low order bits: 0 (000) thru 7 (111)
- Range for n bits $\pm(2^{n-1} - 1)$
 - There are two representations for 0
- Cumbersome addition/subtraction
 - must compare magnitudes to determine sign of result

0 100 = + 4

1 100 = - 4



Signed Numbers: 1's Complement Method

- If N is a positive number, then the negative of N (its 1's complement or N') is

$$N' = (2^n - 1) - N$$

- Example: 1's complement of 7

$$2^4 = 10000$$

$$1 = 00001$$

$$2^4 - 1 = 1111$$

$$7 = 0111$$

$$1000 = -7 \text{ in 1s complement form}$$

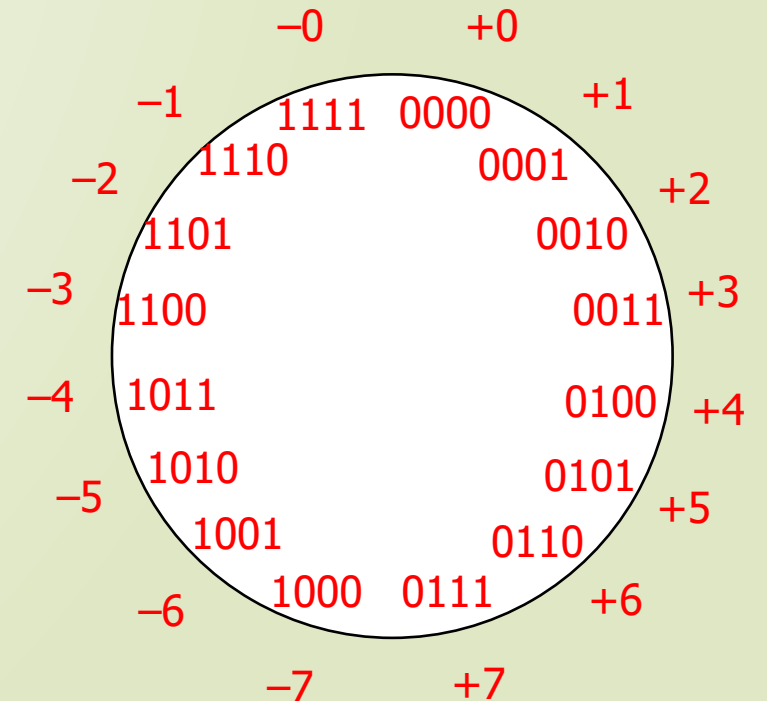
- Shortcut: simply compute bit-wise complement (0111 -> 1000)

Signed Numbers: 1's Complement Method

- Subtraction implemented by 1's complement and then addition
- Two representations of 0
 - Causes some complexities in addition
- High-order bit can act as sign bit

0 100 = + 4

1 011 = - 4

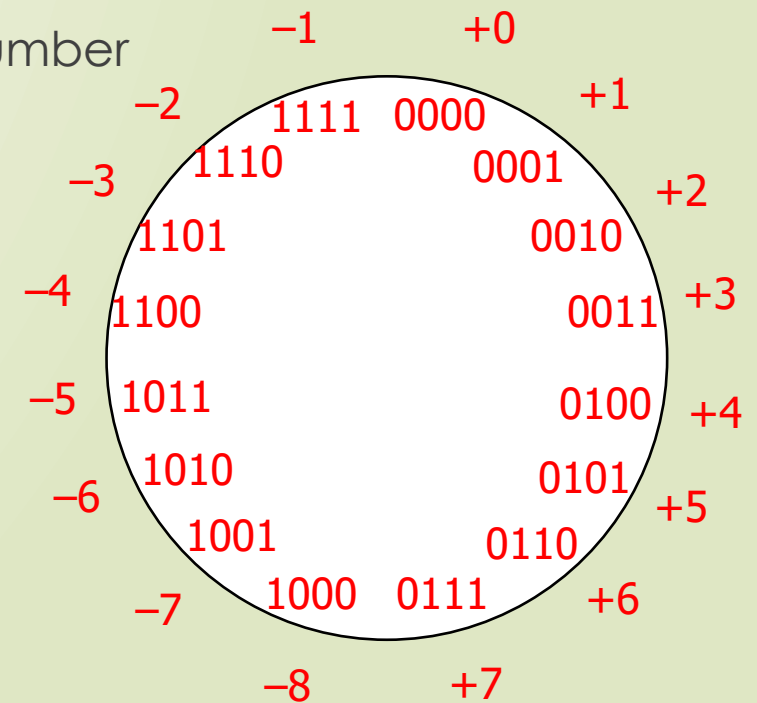


Signed Numbers: 2's Complement Method

- 1's complement with negative numbers shifted one position clockwise
 - Only one representation for 0
 - One more negative number than positive number
 - High-order bit can act as sign bit

0 100 = + 4

1 100 = - 4



Signed Numbers: 2's Complement Method

- If N is a positive number, then the negative of N (its 2's complement or N^*) is

$$N^* = 2^n - N$$

- Example: 2's complement of 7

$$\begin{array}{r} \text{subtract} \quad 2^4 = 10000 \\ \quad \quad \quad 7 = \underline{0111} \\ \hline \quad \quad \quad 1001 = \text{repr. of } -7 \end{array}$$

- Example: 2's complement of -7

$$\begin{array}{r} \text{subtract} \quad 2^4 = 10000 \\ \quad \quad \quad -7 = \underline{1001} \\ \hline \quad \quad \quad 0111 = \text{repr. of } 7 \end{array}$$

- Shortcut: 2's complement = bit-wise complement + 1

- 0111 $\rightarrow$ 1000 + 1 $\rightarrow$ 1001 (representation of -7)
- 1001 $\rightarrow$ 0110 + 1 $\rightarrow$ 0111 (representation of 7)

Summary:

Binary, Signed Integer Representations

B				Values represented		
$b_3b_2b_1b_0$				Sign and magnitude	1's complement	2's complement
0	1	1	1	+ 7	+ 7	+ 7
0	1	1	0	+ 6	+ 6	+ 6
0	1	0	1	+ 5	+ 5	+ 5
0	1	0	0	+ 4	+ 4	+ 4
0	0	1	1	+ 3	+ 3	+ 3
0	0	1	0	+ 2	+ 2	+ 2
0	0	0	1	+ 1	+ 1	+ 1
0	0	0	0	+ 0	+ 0	+ 0
1	0	0	0	- 0	- 7	- 8
1	0	0	1	- 1	- 6	- 7
1	0	1	0	- 2	- 5	- 6
1	0	1	1	- 3	- 4	- 5
1	1	0	0	- 4	- 3	- 4
1	1	0	1	- 5	- 2	- 3
1	1	1	0	- 6	- 1	- 2
1	1	1	1	- 7	- 0	- 1



Question 3

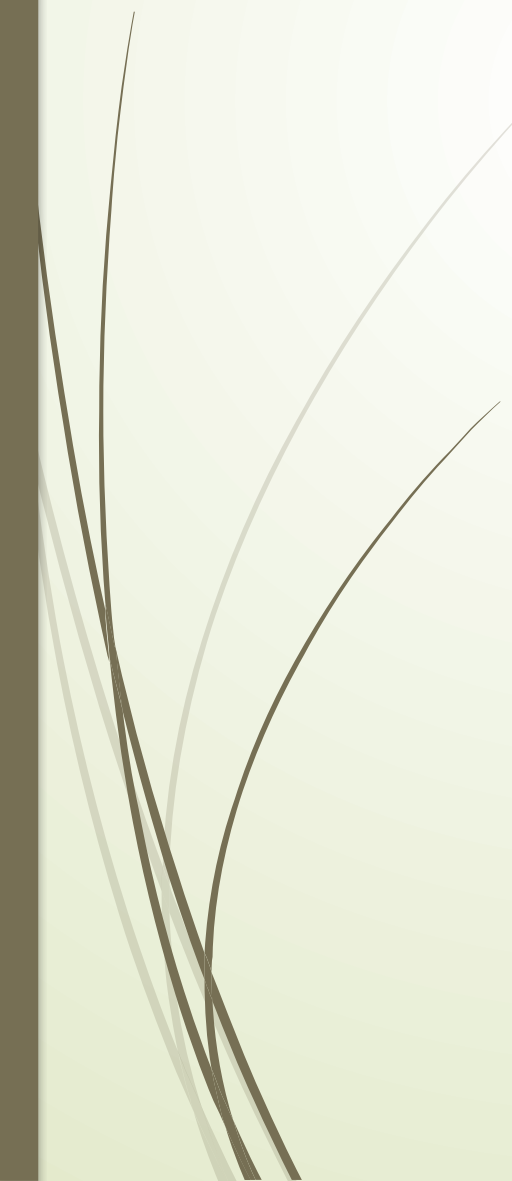
► Convert the binary number

1010 0100 1001 0010 0100 1001 0010 0100₂

to decimal, if the binary is

Unsigned?

2's complement?



Question 3.1

- Converting binary (**unsigned**) to decimal

1010 0100 1001 0010 0100 1001 0010 0100₂

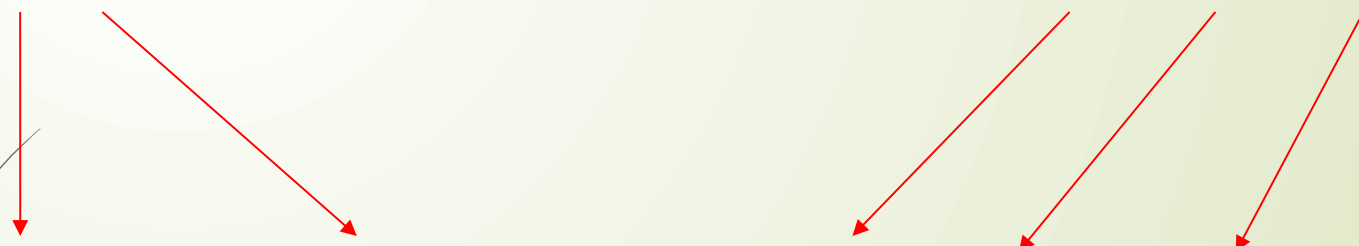


$1*2^{31} + 1*2^{29} + \dots + 1*2^8 + 1*2^5 + 1*2^2$
= 2761050404

Question 3.2

- Converting binary (**2's complement**) to decimal

1010 0100 1001 0010 0100 1001 0010 0100₂


$$\begin{aligned} & -1 * 2^{31} + 1 * 2^{29} + \dots + 1 * 2^8 + 1 * 2^5 + 1 * 2^2 \\ & = -1533916892 \end{aligned}$$

Sign Extension

- Suppose I have a 4-bit 2's complement number and I want to make it into an 8-bit number
 - Positive number
 - add 0's to LHS e.g. 0111 -> 00000111
 - Negative number
 - add 1's to LHS e.g. 1010 -> 11111010

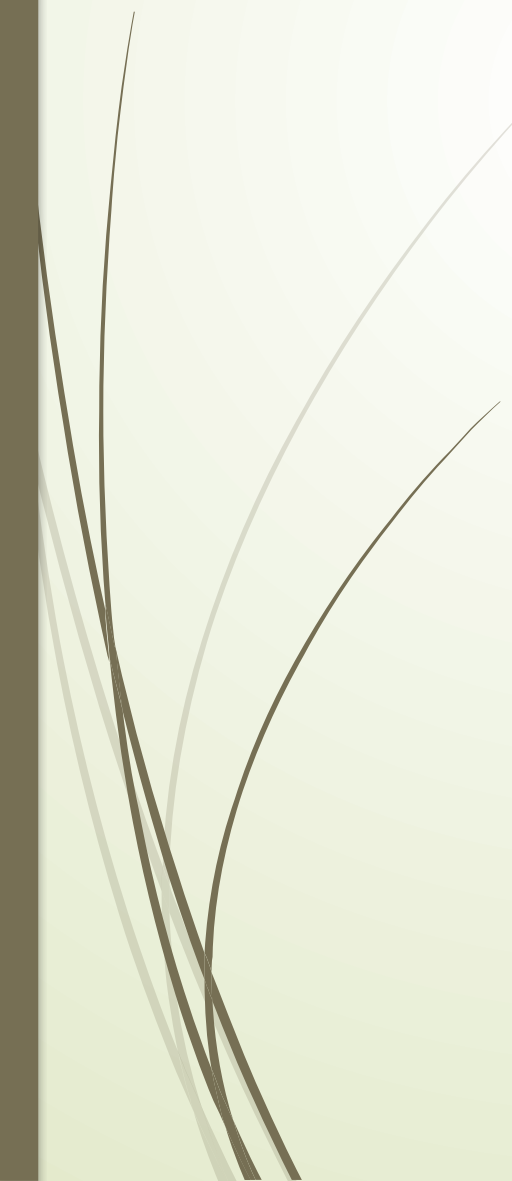


Question 4

➤ Add 0100 1111 and 1001, if they are

Unsigned?

2's complement?



Question 4.1

- Add 0100 1111 and 1001, if they are unsigned
 - Extend 1001 into an 8-bit number: 0000 1001
 - Add 0100 1111
 - 0000 1001
 - 0101 1000

Question 4.1

➤ Add 0100 1111 and 1001, if they are 2's complement

➤ Extend 1001 into an 8-bit number: 1111 1001

➤ Add 0100 1111

➤ 1111 1001

➤ 0100 1000